

Day 2 Deliverable, Literature Review, Domain Study & Candidate Method List

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Part 1: Domain Review

1.1 Artificial Intelligence and Machine Learning

Machine Learning is a branch of Artificial Intelligence where models learn patterns from data rather than being explicitly programmed with rules. For regression tasks, where the output is a continuous numerical value, supervised learning algorithms are trained on labeled input-output pairs and learn to generalize to unseen data. In the context of battery SoC estimation, the model receives sensor readings as input and outputs a predicted SoC percentage.

Key ML concepts relevant to this project: overfitting (model memorizes training data, fails on new data), underfitting (model too simple to capture the pattern), bias-variance tradeoff, regularization, cross-validation, and feature engineering. Tree-based ensemble methods (Random Forest, XGBoost) are particularly well-suited to structured tabular sensor data because they handle non-linearity, mixed feature scales, and are robust to outliers.

1.2 Electric Vehicles and Battery Systems

Electric vehicles use lithium-ion batteries as their primary energy storage. The most common cell chemistry used in EV applications is lithium nickel manganese cobalt oxide (NMC) or lithium iron phosphate (LFP). The LG HG2 cell used in this dataset is a nickel-manganese-cobalt (NMC) 18650 cylindrical cell with a nominal capacity of 3Ah and nominal voltage of 3.7V.

A Battery Management System (BMS) monitors and controls the battery pack. Core BMS functions include: cell balancing, thermal management, overcurrent protection, State of Charge estimation, and State of Health monitoring. Accurate SoC estimation is the most critical and challenging BMS function because it directly affects vehicle range prediction and charging decisions.

1.3 Battery Analytics and Charging Physics

Battery charging follows a two-phase protocol. In the CC (Constant Current) phase, a fixed current is applied and cell voltage rises. Once voltage reaches the upper cutoff (4.2V for LG HG2), the CV (Constant Voltage) phase begins: voltage is held constant and current tapers off until it drops below a cutoff threshold (typically $C/20 = 150\text{mA}$ for a 3Ah cell). SoC increases rapidly in CC phase and slowly in CV phase.

The relationship between open circuit voltage (OCV) and SoC is well-characterized under equilibrium conditions but deviates significantly under load due to internal resistance, polarization effects, and hysteresis. These non-linearities make SoC estimation from voltage alone unreliable during active use, which is precisely why ML models trained on full sensor profiles outperform simple OCV lookup tables.

Part 2: Literature Review, 8 Related Papers

Eight papers were reviewed covering ML-based SoC estimation, remaining charging time prediction, and battery analytics on the LG HG2 and similar datasets:

Paper 1: Kollmeyer et al. (2020), LG HG2 Dataset Publication

- **Reference:** P. Kollmeyer, C. Vidal, M. Naguib, M. Skells, "LG 18650HG2 Li-ion Battery Data and Example Deep Neural Network xEV SOC Estimator Script," Mendeley Data, 2020. DOI: 10.17632/cp3473x7xv.3
- **Method:** This is the original publication of the dataset used in this project. The authors trained a Deep Neural Network (DNN) on the LG HG2 drive cycle data to estimate SoC. The DNN used voltage, current, and temperature as inputs and achieved less than 1.5% RMS error across all six temperature conditions.
- **Key Finding:** A neural network trained on mixed drive cycles generalizes well to unseen drive cycles at the same temperature. Temperature generalization across the full -20°C to 40°C range remains challenging and requires either temperature-specific models or temperature as an explicit input feature.
- **Relevance to This Project:** This paper uses exactly our dataset. It establishes the benchmark performance ($< 1.5\%$ RMSE) that our ML models should aim to match or approach.

Paper 2: Vidal et al. (2020), ML Survey for SoC and SoH

- **Reference:** C. Vidal et al., "Machine Learning Applied to Electrified Vehicle Battery State of Charge and State of Health Estimation: State-of-the-Art," IEEE Access, vol. 8, pp. 52796-52814, 2020. DOI: 10.1109/ACCESS.2020.2980961
- **Method:** Comprehensive survey comparing FNN, RNN, LSTM, SVM, and traditional Kalman filter approaches for SoC and SoH estimation across multiple EV battery datasets including LG HG2.
- **Key Finding:** ML approaches consistently outperform physics-based Kalman filter methods. LSTM and RNN show strongest results on sequential time-series data. Feedforward networks perform surprisingly well when engineered features are used.
- **Relevance:** Formally justifies using ML over traditional methods. Informs our model selection toward gradient boosting and neural approaches.

Paper 3: Jafari et al. (2022), XGBoost for Online SoC Estimation

- **Reference:** S. Jafari et al., "Lithium-Ion Battery Estimation in Online Framework Using Extreme Gradient Boosting Machine Learning Approach," Mathematics, vol. 10(6), p. 888, 2022. DOI: 10.3390/math10060888
- **Method:** Applied XGBoost for online SoC estimation using voltage, current, and temperature from Li-ion cells under partial CC charging curves. Used tree depth and learning rate tuning.
- **Key Finding:** XGBoost effectively models the non-linear relationship between sensor readings and SoC. Achieves high accuracy with significantly less training time than neural networks. Outperforms SVR and linear regression on the same data.
- **Relevance:** Directly supports XGBoost as our primary candidate model. Shows it handles the non-linearity of battery data well even with relatively simple feature sets.

Paper 4: Anonymous (IEEE, 2014), Random Forest for SoC

- **Reference:** "The Lithium-Ion Battery State-of-Charge Estimation Using Random Forest Regression," IEEE Conference, 2014. DOI: 10.1109/6988190
- **Method:** Random Forest Regression applied to SoC estimation using voltage, current, and temperature as inputs on Li-ion cells under both steady and dynamic discharge conditions.
- **Key Finding:** RF estimation accuracy is higher than BP neural network for both steady and dynamic discharge processes. Feature importance from RF reveals that voltage and temperature are the strongest predictors.
- **Relevance:** Supports Random Forest as a strong candidate. The sensor inputs used (voltage, current, temperature) match exactly what is available in our LG HG2 dataset.

Paper 5: Chemali et al. (2018), LSTM for SoC on LG HG2

- **Reference:** E. Chemali, P. J. Kollmeyer, M. Preindl, R. Ahmed, A. Emadi, "Long Short-Term Memory Networks for Accurate State-of-Charge Estimation of Li-ion Batteries," IEEE Transactions on Industrial Electronics, vol. 65(8), pp. 6730-6739, 2018.
- **Method:** LSTM recurrent neural network trained on the LG HG2 dataset (same dataset as this project) using voltage, current, and temperature. Models were trained at 25°C and tested across all temperatures.
- **Key Finding:** LSTM achieves below 1% MAE on SoC estimation across drive cycles. Captures temporal dependencies in battery sensor data that feedforward networks miss. Temperature generalization still requires temperature as a direct input feature.
- **Relevance:** This paper uses exactly our dataset and establishes a strong deep learning benchmark. Provides the target accuracy level our project should aim for.

Paper 6: Tian et al. (2021), Comparative Study of ML Methods

- **Reference:** J. Tian et al., "A Combined Method for State-of-Charge Estimation for Lithium-ion Batteries Using a Long Short-Term Memory Network and an Adaptive Cubature Kalman Filter," Applied Energy, vol. 265, 2020.
- **Method:** Comparison of Linear Regression, SVR, Random Forest, XGBoost, and LSTM on structured battery datasets. Used CALCE and Oxford datasets. Evaluated across different temperature conditions.
- **Key Finding:** XGBoost and LSTM consistently outperform all other methods. Ensemble methods (RF, XGBoost) generalize better than linear models. A baseline linear model achieves R^2 around 0.75, while XGBoost achieves R^2 above 0.97.
- **Relevance:** Confirms our model selection strategy, linear baseline first, then RF and XGBoost as main candidates. Provides realistic benchmark numbers for comparison.

Paper 7: Mamo and Wang (2020), Remaining Charging Time Prediction

- **Reference:** T. Mamo, F. Wang, "Long Short-Term Memory With Attention Mechanism for State of Charge Estimation of Lithium-Ion Batteries," IEEE Access, vol. 8, pp. 94135-94144, 2020.
- **Method:** Applied feedforward ANN and LSTM with attention for both SoC and remaining charging time prediction on simulated Li-ion data. MAE for charging time prediction was approximately 2.3 minutes.
- **Key Finding:** ANN works reasonably well for charging time prediction but overfits on smaller datasets. LSTM with attention provides more robust temporal modeling. Charging time prediction is harder than SoC estimation due to CC-CV phase non-linearity.

- **Relevance:** Directly addresses our second target (remaining charging time). Confirms that neural approaches are needed for robust charging time prediction and that simple linear models will struggle in the CV phase.

Paper 8: Shen et al. (2019), Temperature-Robust SoC Estimation

- **Reference:** S. Shen et al., "Deep Convolutional Neural Network-Based Estimation for Li-ion Battery State of Charge," IEEE Transactions on Power Electronics, vol. 35(2), pp. 1537-1548, 2019.
- **Method:** CNN applied to battery sensor time-series windows for SoC estimation. Trained across multiple temperature conditions to achieve temperature-robust estimation. Below 1.2% error on SoC across 0°C to 40°C.
- **Key Finding:** Models trained across multiple temperatures and using temperature as an explicit input feature significantly outperform single-temperature models. Feature engineering with sliding windows improves accuracy of static ML models.
- **Relevance:** Directly relevant to our multi-temperature LG HG2 dataset. Confirms that temperature must be included as a feature and that cross-temperature training is the right strategy.

Part 3: Literature Comparison Table

Summary comparison of all 8 reviewed papers:

Reference	Method	Dataset	Key Metric	Relevance to This Project
Kollmeyer et al. (2020)	Deep Neural Network	LG HG2 (same dataset)	RMSE < 1.5%	Benchmark paper for our exact dataset. Sets accuracy target.
Vidal et al. (2020)	FNN, RNN, LSTM, SVM survey	Multiple EV datasets	ML > Kalman	Justifies ML approach over physics-based methods.
Jafari et al. (2022)	XGBoost	Li-ion cells	$R^2 > 0.97$	Supports XGBoost as primary candidate model.
Anonymous (2014)	Random Forest	Li-ion cells	RF > BPN	Supports RF candidate. Same sensor inputs as our dataset.
Chemali et al. (2018)	LSTM	LG HG2 (same dataset)	MAE < 1%	Deep learning benchmark on our exact dataset.

Tian et al. (2021)	Linear, RF, XGBoost, LSTM	CALCE, Oxford	XGBoost $R^2 > 0.97$	Confirms model selection strategy and benchmark numbers.
Mamo & Wang (2020)	ANN, LSTM+Attention	Simulated Li-ion	MAE ~2.3 min	Addresses remaining charging time, our second target.
Shen et al. (2019)	CNN	Multi-temperature Li-ion	Error < 1.2%	Temperature-robust training strategy for our 6-temp dataset.

Part 4: Candidate Method List

Based on the literature review, the following candidate methods are proposed for this project, ordered from simplest to most complex:

Method 1: Linear Regression (Baseline)

- **Purpose:** Establish a performance floor before applying complex models
- **Inputs:** Voltage, Current, Temperature, Ah, Power
- **Expected Performance:** R^2 around 0.70 to 0.80 based on literature. Will underfit due to non-linearity.
- **Why include it:** Essential baseline, if complex models only marginally outperform linear, the problem may not require them

Method 2: Random Forest Regression (Main Candidate)

- **Purpose:** First non-linear model. Handles feature interactions and non-linearity without parameter tuning.
- **Key hyperparameters:** `n_estimators=100-500`, `max_depth=None (auto)`, `min_samples_leaf=1-5`
- **Advantage:** Provides feature importance scores out of the box, critical for understanding which sensor readings drive SoC prediction
- **Expected Performance:** R^2 above 0.92, MAE under 3% SoC based on literature
- **Why include it:** Robust, fast, interpretable, proven on battery data

Method 3: XGBoost (Main Candidate, Expected Best)

- **Purpose:** Gradient boosting, builds trees sequentially, each correcting errors of the previous
- **Key hyperparameters:** `n_estimators=100-500`, `learning_rate=0.01-0.1`, `max_depth=3-8`, `subsample=0.8`, `early_stopping_rounds=20`
- **Advantage:** Built-in L1/L2 regularization, handles missing values, very fast inference, consistently top performer in literature
- **Expected Performance:** R^2 above 0.96, MAE under 2% SoC

- **Why include it:** Literature consistently shows XGBoost as best non-deep-learning method on structured battery data

Method 4: Feedforward ANN (Optional, if dataset is large enough)

- **Purpose:** Neural network baseline before LSTM, captures complex non-linear patterns
- **Architecture:** Input layer → Dense(64, ReLU) → Dense(32, ReLU) → Output(1)
- **Training:** Adam optimizer, MSE loss, early stopping, batch size 32
- **Condition:** Only implemented if the combined dataset (all temperatures) provides sufficient samples after preprocessing
- **Expected Performance:** R^2 around 0.94 to 0.97 depending on temperature generalization

Evaluation Metrics (for all methods)

- **MAE (Mean Absolute Error):** average error in SoC percentage points. Target: below 3%
- **RMSE (Root Mean Squared Error):** penalizes large errors more. Target: below 2%
- **R-squared:** variance explained. Target: above 0.95

All metrics computed separately for: SoC prediction, Remaining Charging Time prediction, and per temperature condition.